

ANLP Project Mid Submission: Generating and Evaluating 3D Relevancy Maps for Text Queries

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Abstract

Grounding open-vocabulary natural language queries within 3D scenes is a critical challenge for advancing embodied AI and immersive computing. This project addresses this challenge by focusing on the generation of dense 3D relevancy maps, which assign a relevance score to every point in a 3D point cloud based on a text query. We propose a modular framework that leverages the semantic power of pre-trained 2D Vision-Language Models (VLMs). Our approach first generates 2D relevancy maps from multiple viewpoints and then geometrically 'lifts' and fuses this information into a coherent 3D representation. This framework is compared with state-of-the-art methods, including the implicit, end-to-end LERF and the explicit, decoupled ConceptFusion. A comparative analysis is conducted by probing these models with a structured set of queries from simple object descriptions to complex spatial and abstract reasoning tasks to evaluate their representational capabilities. The goal is to analyze the architectural trade-offs and validate the hypothesis that a modular, VLM-driven approach can offer superior performance and flexibility, particularly for complex, human-centric queries.

1 Introduction

1.1 Project Overview

The advancement of artificial intelligence, particularly in the domains of embodied agents such as robotics and immersive computing platforms like augmented and virtual reality (AR/VR), is fundamentally bottlenecked by the machine's ability to comprehend its physical environment. While significant progress has been made in geometric reconstruction—creating dimensionally accurate 3D models of spaces—a true understanding transcends mere geometry. It requires a deep, semantic interpretation of the scene, enabling interaction based on natural, human-like communication. The next

frontier is to bridge the gap between the unstructured, continuous nature of 3D space and the symbolic, compositional structure of human language.

This project is motivated by the need for systems that can ground open-vocabulary language queries directly into the 3D structure of a scene. Such a capability would unlock a new paradigm of human-AI interaction. For instance, a domestic robot could be instructed to "find the sponge to clean the coffee spill," a task that requires not only recognizing a "sponge" but also understanding its affordance ("to clean") and its spatial context relative to a "spill". Similarly, a maintenance technician using an AR headset could ask the system to "highlight all pipes showing signs of corrosion," a query that involves identifying objects, their material properties, and their state. These examples illustrate a need that goes far beyond recognizing a predefined, closed set of object categories. They demand a system that can interpret and localize a virtually infinite range of objects, attributes, spatial relationships, and even abstract concepts.

To address this challenge, this project focuses on the generation of "3D relevancy maps." A 3D relevancy map is formally defined as a dense, continuous field superimposed upon a 3D scene representation (e.g., a point cloud or the volume of a Neural Radiance Field). In this field, every point in 3D space is assigned a continuous-valued score, typically between 0 and 1, that quantifies its relevance to a given natural language text query. This representation marks a significant departure from traditional 3D object detection, which typically outputs a discrete set of 3D bounding boxes corresponding to a closed set of object classes. A relevancy map is a more general and powerful construct. It can localize not only well-defined objects ("the red chair") but also parts of objects ("the armrest of the chair"), material properties ("the wooden texture on the floor"), and, most importantly, abstract concepts that lack clear geometric boundaries. For example,

a relevancy map can highlight "the most cluttered area on the desk," "a comfortable place to read," or "the region with the most natural light". This ability to ground non-object-centric and abstract language is critical for building truly intelligent systems that can reason about the world in human-centric terms.

1.2 Objectives

The central objective of this project is to architect, implement, and rigorously validate a novel, modular framework that leverages the power of pre-trained Vision-Language Models (VLMs) for generating high-fidelity 3D relevancy maps from open-vocabulary text queries. This framework will be benchmarked against leading state-of-the-art methods to assess its efficacy and identify its unique strengths.

The project is guided by two primary hypotheses:

Primary Hypothesis: A modular architectural approach, which first generates 2D relevancy maps from multiple views and then explicitly "lifts" and fuses them into a 3D representation using geometric constraints, can achieve performance comparable or superior to that of end-to-end implicit methods (like LERF). This modularity is hypothesized to offer greater flexibility, interpretability, and debuggability.

Secondary Hypothesis: The proposed framework, by directly leveraging the compositional understanding of modern VLMs in a decoupled 2D stage, will excel at handling complex queries that involve nuanced spatial relationships ("the book to the left of the lamp") and abstract concepts ("a source of light"), which are challenging for methods that tightly entangle semantic and geometric learning.

This report outlines the theoretical foundations, a review of related work, the proposed methodology, a comprehensive experimental protocol, and a project timeline to investigate these hypotheses.

2 Background

OpenAI’s Contrastive Language-Image Pre-training (CLIP) model(Radford et al., 2021) is the archetypal VLM and serves as the foundational component for both LERF(Kerr et al., 2023) and ConceptFusion(Jatavallabhula et al., 2023), as well as the proposed framework in this report. A detailed understanding of its architecture and training is therefore essential. CLIP consists of

two independent neural network encoders that are trained jointly: an Image Encoder (typically a Vision Transformer or ResNet) and a Text Encoder (a Transformer-based language model). A linear projection layer maps the outputs of both encoders into a common, multi-modal embedding space where images and text can be directly compared. CLIP learns to align these two modalities through a contrastive learning objective on a massive dataset of 400 million (image, text) pairs. During training, the model maximizes the cosine similarity of correct image-text pairs while minimizing the similarity of incorrect pairings. This forces the model to learn a shared representation space where semantically similar concepts are located close to one another. The power of this learned space is realized during zero-shot inference, where the similarity between a query text’s embedding and visual features provides a measure of relevance. While CLIP is trained on 2D images, its robust feature representations are key for 3D semantic understanding. By capturing multiple 2D views of a 3D scene, one can extract CLIP features from each view and then aggregate or distill this information into a coherent 3D semantic representation using known camera poses and geometry.(Radford et al., 2021)

The ability to ground language in 3D space is predicated on the underlying computational representation of that space. Two dominant paradigms exist. Explicit Representations model geometry as a discrete set of elements, like a point cloud—an unordered set of points $P = \{p_i | p_i \in \mathbb{R}^3\}$ —which are the direct output of LiDAR or RGB-D sensors.

Implicit Representations, in contrast, model a scene as a continuous function. The most influential of these is the Neural Radiance Field (NeRF)(Mildenhall et al., 2020), which uses a multilayer perceptron (MLP) to learn a mapping $F_{\Theta} : (\mathbf{x}, \mathbf{d}) \rightarrow (\mathbf{c}, \sigma)$ from a 3D location $\mathbf{x}$ and viewing direction $\mathbf{d}$ to a color $\mathbf{c}$ and volume density σ , enabling photorealistic novel view synthesis. The choice between these is a fundamental architectural decision; ConceptFusion builds upon an explicit point cloud, while LERF integrates its language field into an implicit NeRF.

The last several years have witnessed a paradigm shift in computer vision, moving away from supervised models trained on fixed label sets towards Vision-Language Models (VLMs), which are trained on web-scale, noisy data. This process imbues them with a broad, general-purpose under-

standing of visual concepts and their relationship to natural language, enabling

zero-shot learning: the ability to perform a task without explicit, task-specific training examples. This flexibility is ideal for open-vocabulary scene understanding. This evolution marks a shift from geometric-first to semantic-first 3D understanding. Traditional pipelines focused on geometric reconstruction, with semantic labeling as a secondary, closed-set task. Modern approaches like LERF and ConceptFusion start with a powerful semantic model. LERF distills the VLM’s knowledge

into the geometry, while ConceptFusion uses geometry as a scaffold to

fuse semantic features. The core problem is reframed from "How do we assign labels to this geometry?" to "Where within this geometry does this pre-understood semantic concept exist?" This shift unlocks the ability to ground abstract concepts that are well-represented in the VLM’s training data but may lack a clear geometric definition.

3 Related Work

The task of grounding language in 3D scenes has seen the emergence of several distinct architectural paradigms, broadly categorized into implicit field approaches, explicit mapping approaches, and high-level reasoning frameworks.

Language Embedded Radiance Fields (LERF)(Kerr et al., 2023) pioneer an approach that directly embeds language into an implicit neural representation of a scene. LERF augments a standard NeRF(Mildenhall et al., 2020) by adding a second MLP head that predicts a CLIP feature vector for any 3D point

$\mathbf{x}$ at a given scale s . The model is trained by jointly optimizing the standard NeRF color reconstruction loss and a language loss, which is supervised by volume-rendered CLIP(Radford et al., 2021) embeddings against 2D CLIP features from training views. This effectively "distills" multi-view 2D semantic information into a dense, 3D-consistent language field. A key innovation is the use of a multi-scale feature pyramid, allowing a single 3D point to be associated with concepts at different scales (e.g., "wooden spoon" vs. "cutlery"). To enforce geometric consistency and produce crisper maps, LERF also incorporates a regularization field supervised by features from the DINO model(Caron et al., 2021), which is known for its excellent emergent object segmentation prop-

erties. After training, a LERF can be queried efficiently by computing the cosine similarity between a text prompt’s embedding and the language field’s output across the entire 3D volume.

ConceptFusion(Jatavallabhula et al., 2023) takes a fundamentally different, decoupled approach. It leverages existing technologies, first using a standard SLAM system to build an explicit 3D point cloud map from an RGB-D video stream. In parallel, it extracts dense, pixel-aligned semantic features from each 2D frame using a foundation model like CLIP. The core innovation is its method for

zero-shot pixel alignment, which generates pixel-level features from global-level models like CLIP. It uses a class-agnostic instance segmentation model to identify regions, computes "local" CLIP features for each region and a "global" feature for the whole image, and then intelligently fuses them to produce a final feature vector for each pixel. These dense 2D feature maps are then "lifted" into 3D by projecting them onto the SLAM-generated geometry using known camera poses and depth. Features from multiple views are fused over time to create a robust, unified 3D semantic map where each point is annotated with a rich feature vector.

While LERF and ConceptFusion focus on dense relevancy maps, other models aim for higher-level reasoning. LLaVA (Large Language and Vision Assistant)(Liu et al., 2023) is a pioneering multimodal conversational agent that connects a pre-trained CLIP vision encoder to a Large Language Model (LLM) like Vicuna via a simple projection matrix. It is fine-tuned on a large dataset of instruction-following examples generated by GPT-4 from 2D images, enabling it to hold coherent conversations about visual content but without any innate 3D understanding. SceneGPT(Chandhok, 2024) bridges this gap by introducing an abstraction layer. It first processes a 3D scene into a structured, language-readable

3D Scene Graph (3DSG), typically a JSON object describing objects, their properties, and spatial relationships. This structured text is then passed into an LLM’s context window, allowing it to parse the representation and answer complex spatial and functional queries. SceneGPT(Chandhok, 2024) excels at high-level reasoning but does not produce a dense relevancy map.

The distinct philosophies of these models lead to a trade-off between dense, low-level grounding and sparse, high-level reasoning. The progression from LERF/ConceptFusion to LLaVA/SceneGPT illus-

trates an evolution from localization to understanding. The former systems answer "Where is concept X?", while the latter answer "What about concept X?". This suggests a powerful future architecture would likely integrate both capabilities, using a dense grounding system as a "perceptual front-end" for an LLM-based "reasoning back-end."

4 Dataset Details

We evaluate our models using three sources of 3D scene data: the LERF dataset, the LERF-TOGO dataset, and a custom dataset constructed from CAD models using BlenderNeRF. This combination enables testing open-vocabulary object localization, task-oriented grasping, and controlled evaluation of spatial reasoning.

4.1 LERF Dataset



Figure 1: Sample scene from the LERF (Kerr et al., 2023) dataset showing a variety of household objects.

The LERF dataset contains scenes with different types of objects, supporting open-vocabulary queries and instance-level semantic localization. It allows testing the model’s ability to identify and localize multiple objects in complex arrangements.

4.2 LERF-TOGO Dataset



Figure 2: NeRF-reconstructed point cloud from LERF-TOGO(Rashid et al., 2023) showing five mugs of different colors

LERF-TOGO provides scenes suitable for zero-shot, task-oriented grasping. It is annotated for object-part queries, e.g., “handle of the mug” or “rim of the cup,” enabling evaluation of models on semantic grasping and functional affordance recognition.

4.3 Custom Dataset Pipeline



Figure 3: Top-down view of a NeRF-reconstructed point cloud from our custom dataset.

We construct our custom dataset using publicly available CAD models, such as apartment floor plans, rendered with BlenderNeRF(Eraafat, 2023) . Multi-view RGB images are generated from predefined camera poses, and the corresponding extrinsic and intrinsic camera parameters are stored in a format compatible with LERF. This setup allows systematic design of scenes with diverse spatial arrangements, placing objects above, below, or beside each other, and enables targeted queries to evaluate spatial reasoning, relative placement, and compositional understanding beyond simple object retrieval.

5 Proposed Methodology

Earlier, we had proposed an approach for the task of getting relevancy maps for general text queries that uses LERF as the main model. Basically, our contribution is to map general text queries to text keywords, which can be directly used for LERF.(Kerr et al., 2023) We have completed this approach by implementing LLAVA (Liu et al., 2023) model. Since there are limitations of the CLIP(Eraafat, 2023) model, which is internally used by LERF, we have proposed another entirely VLM-based approach, which does not use CLIP as follows.

5.1 VLM-based approach

This section presents the architecture of the proposed novel framework, termed Lift-Fuse-Map (LFM). The LFM framework is designed with a

modular philosophy, explicitly separating the task of semantic interpretation from the task of geometric aggregation. This design choice aims to harness the power of state-of-the-art 2D VLMs while maintaining a robust and interpretable geometric pipeline. The LFM framework operates as a three-stage pipeline. It takes as input a sequence of RGB-D images with their corresponding camera poses and a natural language text query. The final output is a dense 3D point cloud of the scene, where each point is assigned a relevancy score corresponding to the input query. This output can be easily visualized as a 3D heatmap. The explicit decoupling of stages is a key strategic advantage. In contrast to end-to-end models like LERF where semantics and geometry are learned jointly, the LFM architecture isolates potential sources of error. This modularity makes the system highly debuggable and allows for independent upgrades of its components.

The first stage is responsible for all semantic processing, operating purely in the 2D domain. A powerful VLM, which is different than CLIP (Radford et al., 2021), is chosen as the semantic backbone. For each RGB image, it is passed through the VLM’s vision encoder to produce a dense feature map. The input text query is passed through the VLM’s text encoder to obtain a single query embedding. A 2D relevancy map is then generated by computing the cosine similarity between the text query embedding and the visual feature vector at every spatial location. This process is repeated for every image in the sequence.

The second stage performs the critical geometric transformation, lifting the 2D semantic information into the 3D world space. First, a unified 3D point cloud of the entire scene is constructed using the full sequence of RGB-D frames and their associated camera poses. The core of the lifting process involves back-projecting the 2D relevancy scores onto this 3D point cloud. For each pixel

(u, v) with a score $R(u, v)$, its corresponding depth value $d(u, v)$ is used with the camera’s known intrinsic matrix K and extrinsic pose matrix T to calculate the 3D world coordinates $\mathbf{P}_{world}$. The transformation is given by the equation:

$$\mathbf{P}_{world} = T^{-1} \left(d(u, v) \cdot K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \right)$$

Each point in the global 3D cloud corresponding to a pixel in this view is then assigned the relevancy score $R(u, v)$.

A single viewpoint provides an incomplete and potentially noisy signal. The final stage aggregates the information from all views to produce a coherent and robust final map. The lifting process is performed for every view, meaning a single 3D point may be assigned multiple relevancy scores. A fusion algorithm, such as computing the mean or a confidence-weighted average, is applied to combine these scores into a single value for each point. After fusion, a final spatial denoising step (e.g., a 3D Gaussian blur or bilateral filter) is applied to the point cloud to remove isolated spurious activations, producing the final, clean 3D relevancy map.

5.2 Experimental Protocol for Comparative Probing

To rigorously evaluate the proposed LFM framework against the state-of-the-art, a comprehensive and fair experimental protocol is required. This protocol is designed to probe the models’ capabilities across a wide range of linguistic complexities on a consistent set of 3D scenes.

The foundation of a fair comparison is the correct preparation and deployment of the baseline models, LERF and ConceptFusion. The official open-source implementations of both models will be used on the same pre-processed ScanNet scenes. As correctly noted, both models provide a mechanism for direct querying of their internal 3D language representations. For LERF, the optimized network’s language field will be queried by sampling a dense grid of 3D points, getting their CLIP embeddings, and computing the cosine similarity with the query text’s embedding. For ConceptFusion, the relevancy score for each point in the final 3D point cloud is the cosine similarity between its stored feature vector and the query text’s embedding. This ensures that the output from all three models is a comparable 3D relevancy map.

A meticulously designed set of textual queries is essential for a nuanced evaluation. The query set will be structured into four categories of increasing

complexity, drawing inspiration from diagnostic datasets like ViGiL3D. A total of 80 unique queries will be crafted.

Category 1: Simple Object Presence (20 queries):

These queries test the fundamental ability to identify and localize common, well-defined objects.

Examples: "a chair," "the television," "a pillow," "the keyboard."

Category 2: Fine-grained and Attribute-based (20 queries): These queries assess specificity and the ability to ground visual attributes like color, material, and texture.

Examples: "a wooden chair," "the blue pillow," "a metal lamp," "the texture on the brick wall."

Category 3: Spatial Relationships (20 queries):

These queries probe compositional understanding and the ability to reason about the relative positions of objects.

Examples: "the trash can under the desk," "a book to the left of the monitor," "the plant on the windowsill," "the object between the two speakers."

Category 4: Complex and Abstract Concepts (20 queries): This category tests the limits of the models by querying for affordances, abstract properties, and concepts that are not simple objects.

Examples: "something you can sit on," "a source of light," "the most cluttered area in the room," "a place to put my keys."

6 Evaluation Plan

6.1 Quantitative Metrics

3D Intersection over Union (3D IoU): For queries in Categories 1 and 2 that refer to specific, localizable objects, 3D IoU is the gold-standard metric for localization accuracy. The process is as follows:

1. The continuous 3D relevancy map is thresholded at an appropriate level (e.g., 0.5) to produce a binary segmentation of the 3D space into "relevant" and "not relevant" regions.
2. The 3D IoU is calculated as the volume of the intersection between this predicted relevant region and the ground-truth 3D bounding box of the target object, divided by the volume of their union:

$$\text{IoU} = \frac{\text{Volume}(\text{Predicted} \cap \text{GroundTruth})}{\text{Volume}(\text{Predicted} \cup \text{GroundTruth})} \quad (1)$$

Grounding Accuracy (Acc@IoU): Following standard practice in the visual grounding literature, a grounding attempt is considered a "success" if the 3D IoU between the predicted region and the ground-truth object exceeds a certain threshold. We report accuracy at two common thresholds: Acc@0.25 IoU and Acc@0.5 IoU.

Precision and Recall for Abstract Concepts:

For queries in Categories 3 and 4, where ground-truth bounding boxes are often ill-defined or non-existent, a different quantitative approach is needed. For a subset of these queries, a sparse set of 3D points will be manually annotated as the "ground-truth relevant set." The evaluation can then be framed as an information retrieval task:

- All points in the scene's point cloud are ranked by their relevancy score from the model.
- Precision is computed as the fraction of retrieved points that are in the ground-truth set.
- Recall is computed as the fraction of ground-truth points that are retrieved at different retrieval depths (k).

This provides a numerical measure of performance even for abstract queries.

6.2 Qualitative Assessment

Quantitative metrics can sometimes be misleading and do not capture the full behavior of a model. Therefore, a thorough qualitative analysis will be conducted.

Visual Inspection of Heatmaps: For a representative subset of queries from all four categories, the generated 3D relevancy heatmaps from LFM, LERF, and ConceptFusion will be rendered side-by-side. These visualizations will be critically examined to assess:

- **Specificity:** How tightly does the relevancy map conform to the target object or region? Are there significant "bleeds" into neighboring objects or the background?

- **Completeness:** Is the entire extent of the relevant object or region highlighted, or does the model only focus on a small, discriminative part?
- **Spurious Activations:** Does the model produce high-relevancy scores in completely irrelevant parts of the scene (false positives)?

Failure Mode Analysis: A systematic analysis of the cases where each model fails will be performed. Failures will be categorized to identify systemic weaknesses. For example:

- Does a model consistently fail on small objects?
- Is it sensitive to occlusion?
- Does it confuse objects with similar colors or shapes?
- Does it fail to parse complex spatial prepositions correctly?

This analysis is crucial for understanding the underlying limitations and behavior of each model.

7 Progress to Date

7.1 Enhancing LERF for General Queries

LERF effectively generates relevancy maps for objects but cannot handle general queries as it only detects object-level relevance. To address this limitation, we developed a two-step pipeline that converts general queries into object descriptions compatible with LERF.

Step 1: Rendering Scenes from NeRF Point Clouds:

We render RGB images from NeRF-reconstructed point clouds of the scene using predefined camera poses looking at the scene. This produces images that capture the scene from a controlled viewpoint for subsequent processing.



Figure 4: RGB image rendered from a point cloud

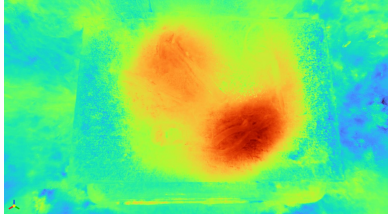
Step 2: Object Description via LLaVA: The rendered images are passed to LLaVA along with a general query. LLaVA outputs concise 2–3 word object descriptions, which LERF can interpret. Example mappings include:

- *The object you would pick to hang a picture, not the one that loosens screws* → Hammer
- *Used to bend a wire or remove a small nail* → Pliers
- *Which tool can I use to rotate screws and fix them in wall* → Screwdriver

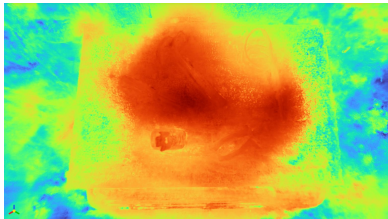
Experimental Example To illustrate this pipeline, we consider a scene containing a hammer, screwdriver, soldering iron, measuring tape, and pliers. All images are rendered for direct comparison.



(a) RGB point cloud image of the scene.



(b) LERF relevancy map for object query "Pliers".



(c) LERF relevancy map for task-based query "Used to bend a wire or remove a small nail".

Figure 5: Comparison of the RGB point cloud and LERF relevancy maps from the same viewpoint. Task-based queries are translated to object queries via LLaVA.

As seen in Figure 5, LERF produces accurate relevancy maps for simple object queries (Fig. 5b). For task-based queries that LERF cannot handle directly (Fig. 5c), LLaVA correctly identifies the target object ("Pliers") from the instruction. LERF then generates a relevancy map over the pliers, successfully grounding complex instructions into localizable objects.

This pipeline extends LERF's capabilities to handle general text instructions and enables evaluation of object localization in task-oriented scenarios.

8 Tentative timeline

8.1 Week 1: Foundations and Baselines

This week will focus on setting up the computational environment, acquiring and pre-processing the selected ScanNet scenes, and deploying the baseline models. The goal is to establish a repeatable workflow for generating 3D relevancy maps from LERF and ConceptFusion on a test scene.

8.2 Week 2: Development of the LFM Framework

The focus of this week is the end-to-end implementation of the proposed LFM framework. This includes developing the modules for generating 2D relevancy maps (Stage 1), lifting them to the 3D point cloud (Stage 2), and implementing the multi-view fusion and denoising strategies (Stage 3).

8.3 Week 3: Execution of Comparative Experiments

This week is dedicated to running the full-scale experimental protocol. The LERF, ConceptFusion, and LFM models will be run on all selected scenes against the entire 80-query set. All resulting 3D relevancy maps and visualizations will be systematically logged and stored for analysis.

8.4 Week 4: Analysis, Synthesis, and Reporting

The final week will involve comprehensive analysis of the experimental results. This includes computing all quantitative metrics, conducting an in-depth qualitative review of the visualizations to identify successes and failures, and synthesizing all findings into the final project report.

Team Contributions Pavan led the project by setting up the codebase, conducting experiments, generating results, and developing the main ideas while guiding the overall direction. Aditya formulated precise research questions and led the writing of the report, incorporating contributions from Pavan. Vishnu contributed by reading relevant papers and participating in team discussions.

Code The code for this project is available at https://github.com/codepk37/anlp_group.

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A Additional Experiments: LERF Limitations

We conducted additional qualitative experiments to probe the reasoning capabilities of LERF. The examples below highlight cases where the model struggles with queries involving spatial or comparative reasoning.



Figure 6: Scene used for LERF text-query experiments.

Scene Setup: The reference scene used for all text–query experiments.

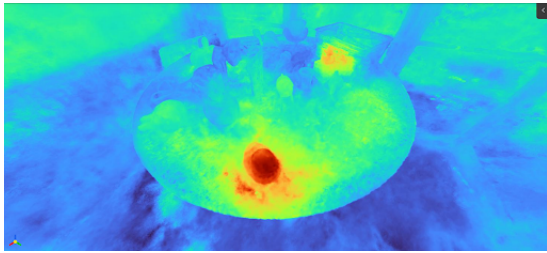


Figure 7: LERF correctly detects both duck instances.

Query: Duck

Observation: LERF correctly highlights both duck instances (in red), demonstrating instance-level object recognition.

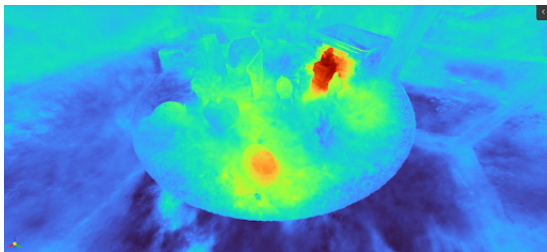


Figure 8: Failure case: LERF cannot reason about spatial relationships.

Query: Duck above Rubik's cube

Observation: Both ducks and the Rubik's cube are highlighted. LERF fails to capture the spatial relationship above; it detects object presence but not relative arrangement.

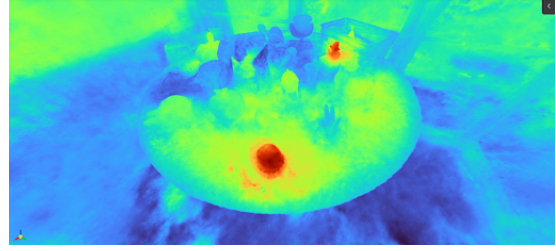


Figure 9: Failure case: LERF cannot reason over comparative size attributes.

Query: Bigger duck

Observation: Both ducks are highlighted equally without distinction. LERF does not reason about relative size attributes between objects.

Future Work: We will also try to address these limitations and improve the model's handling of spatial relationships and comparative attributes.